

NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Civil Engineering
MID SEMESTER EXAMINATION, JAN- JUN 2022

B.Tech

Semester II

Course Name: Surveying and Field Practice

Maximum Time: 2 h

Course Code: CE23103

Max. Marks: 20

Instructions:

1. Attempt all the questions.
2. Assume the data wherever necessary as per standards.

Question No.	Questions	Marks	CO	BL												
1	Differentiate plane surveying over geodetic surveying.	2	1	1,2												
2	Write down about the principle of chain surveying. <i>Triangulation</i>	2	1,2	1,2,3												
3	Write a short note on reciprocal ranging.	2	1,2	2,3,4												
4	A 20m steel tape was standardized on flat ground at a temperature of 20 °C and under a pull of 15 kg. The tape was used in catenary temperature of 30 °C and under a pull of 20 kg. The cross-sectional area of the tape is 0.22 cm ² and its total weight is 400 grams. The Young's Modulus and co-efficient of linear expansion of steel tape are 2.1 kg/cm ² and 11× 10 ⁻⁶ /°C respectively. Find the corrected horizontal distance.	4	1, 2	1,2,3,4												
5	The following are the observed bearing of the lines of a closed traverse ABCDEA with a compass. Find the interior angles. <table border="1"><thead><tr><th>Line</th><th>Fore Bearing</th></tr></thead><tbody><tr><td>AB</td><td>60°30'</td></tr><tr><td>BC</td><td>122°0'</td></tr><tr><td>CD</td><td>46°0'</td></tr><tr><td>DE</td><td>205°30'</td></tr><tr><td>EA</td><td>300°0'</td></tr></tbody></table>	Line	Fore Bearing	AB	60°30'	BC	122°0'	CD	46°0'	DE	205°30'	EA	300°0'	6	1,2	1,2,3,4
Line	Fore Bearing															
AB	60°30'															
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6	Differentiate between whole circle bearing and reduced bearing.	4	1,2	1,2												

